



Computer Engineering Dept.
Faculty of Engineering
Cairo University



Machine Learning Assignment2

Rawan Mostafa Mahmoud

Sec1

BN:18

ID:9210423

Problem 2.16:

$$H = \left\{ h_c(x) = \text{sign} \left(\sum_{i=0}^D c_i x^i \right) \right\}$$

a) $h_c(x) = \text{sign}(C_0 + C_1 x + C_2 x^2 + C_3 x^3 + \dots + C_D x^D)$

$\hookrightarrow D+1$ Coefficients

$\therefore$ we need $D+1$ input points $x_1, x_2, \dots, x_{D+1}$ to solve the above system and get $C_0, C_1, \dots, C_D$

• In Matrix Form: $y = X C$ such that $y \in \{-1, 1\}$
 $C = X^{-1} y$

$$\therefore \text{sign} \left(\sum_{i=0}^D c_i x^i \right) = \text{sign}(y) = y$$

$\therefore H$ shatters $D+1$ points

b) A polynomial of degree D has at most D real roots, which means it can change signs at most D times $\rightarrow$ ①

$\rightarrow$ Counter example:

Try to shatter $D+2$ points

$$y_1 = +1 \quad y_2 = -1 \quad y_3 = +1, \dots, y_{D+2} = (-1)^{D+1}$$

$\hookrightarrow$ These need $D+1$ sign changes !!

But from ① we know that only D sign changes can be reached from degree D polynomial

$\therefore H$ can't shatter $D+2$ points

$\therefore$ VC dimension of H is exactly $(D+1) \#$

Problem 2.24:

a)

$$\bar{g}(x) = E_D(g^D(x))$$

$$h(x) = ax + b$$

$$a = \frac{y_2 - y_1}{x_2 - x_1} = \frac{x_2^2 - x_1^2}{x_2 - x_1} = \frac{(x_2 - x_1)(x_2 + x_1)}{x_2 - x_1} = x_2 + x_1$$

$$b = x_1^2 - ax_1 = x_1^2 - (x_2 + x_1)x_1 = -x_1x_2$$

$$\therefore h(x) = (x_2 + x_1)x - x_1x_2$$

$$\begin{aligned}\bar{g}(x) &= E[(x_2 + x_1)x - x_1x_2] \\ &= 0 \cdot x - 0 = 0\end{aligned}$$

→ Due to Uniform Distr.

$$E_D(x_1) = \frac{1}{2} \int_{-1}^1 x_0 dx = 0$$

$$E_D(x_2) = 0$$

$$E_D(x_1x_2) = E(x_1) \cdot E(x_2) = 0$$

$$d). \text{Bias} = E_x[(\bar{g}(x) - f(x))^2] = E_x[(0 - x^2)^2] = E_x[x^4] \\ = \frac{1}{2} \int_{-1}^1 x^4 = \boxed{\frac{1}{5}} \#$$

$$\bullet \text{Variance} = E_x[E_D[(g^D(x) - \bar{g}(x))^2]]$$

$$= E_x[E_D[(x_1 + x_2)x - x_1x_2 - 0]^2]]$$

$$= E_x[E_D[(x_1 + x_2)^2 x^2 - 2x_1x_2x(x_1 + x_2) + x_1^2x_2^2]]$$

$$= E_x[\underbrace{x^2 E_D(x_1^2 + 2x_1x_2 + x_2^2)}_{I_1} - 2x \underbrace{E_D(x_1x_2(x_1 + x_2))}_{I_2} + \underbrace{E_D(x_1^2x_2^2)}_{I_3}]$$

$$\bullet I_1 = \int_{-1}^1 \int_{-1}^1 \frac{1}{4} (x_1^2 + 2x_1x_2 + x_2^2) dx_1 dx_2$$

$$= \frac{1}{4} \int_{-1}^1 \left[\frac{1}{3} x_1^3 + x_1^2 x_2 + x_2^2 x_1 \right]_{-1}^1 dx_2 = \frac{1}{4} \int_{-1}^1 \frac{2}{3} + 2x_2^2 = \frac{2}{3}$$

$$\bullet I_2 = \frac{1}{4} \int_{-1}^1 \int_{-1}^1 x_1^2 x_2 + x_1 x_2^2 dx_1 dx_2$$

$$= \frac{1}{4} \int_{-1}^1 \left[\frac{1}{3} x_1^3 x_2 + \frac{1}{3} x_2^3 x_1 \right]_{-1}^1 dx_2 = \frac{1}{4} \int_{-1}^1 \frac{2}{3} x_2 + \frac{2}{3} x_2^3 = 0$$

$$\bullet I_3 = \frac{1}{4} \int_{-1}^1 \int_{-1}^1 x_1^2 x_2^2 dx_1 dx_2 =$$

$$= \frac{1}{4} \int_{-1}^1 \left[\frac{1}{3} x_1^3 x_2^2 \right]_{-1}^1 dx_2 = \frac{1}{4} \int_{-1}^1 \frac{2}{3} x_2^2 dx_2 = \frac{1}{9}$$

$$\text{Variance} = E_x\left(\frac{2}{3}x^2 + \frac{1}{9}\right) = \frac{1}{2} \int_{-1}^1 \frac{2}{3}x^2 + \frac{1}{9} dx = \boxed{\frac{1}{3}} \#$$

$$E_{\text{tot}} = \text{Bias} + \text{Variance} = \frac{8}{15} = \boxed{0.5333} \#$$